Customer Feedback Analysis Using Natural Language Processing (NLP)

1. Executive Summary

This report presents the methodology, experimental results, and operational insights for **Day 3** of the **Uber Eats Operational & Customer Intelligence System** capstone project. The primary objective is to analyze customer reviews using Natural Language Processing (NLP) and Machine Learning (ML) techniques to classify sentiment, extract frequent dissatisfaction terms, and evaluate operational feedback.

Using a dataset of 1,984 customer review records, an end-to-end NLP framework was built. Text preprocessing steps included lowercased standardization, punctuation and URL removal, stop-word filtering with explicit preservation of negation terms (e.g., "not", "no"), and WordNet lemmatization. Feature extraction was performed using Term Frequency–Inverse Document Frequency (TF-IDF) with unigram and bigram representations. Two supervised classification models—Multinomial Naive Bayes and Linear Support Vector Machine (SVM)—were trained and evaluated against a lexicon-based baseline (VADER).

2. Business Context and Objectives

2.1 Business Problem in Food Delivery

In multi-sided food delivery platforms like Uber Eats, customer reviews and star ratings serve as vital signals for service quality, customer retention, and operational health. Negative feedback often spans multiple operational vectors:

1. **Delivery Operations:** Late deliveries, cold food due to extended transit times, or courteousness of delivery partners.

2. **Restaurant Operations:** Missing items, wrong order preparation, poor food packaging, or substandard meal quality.
3. **Platform & App Experience:** Payment errors, order tracking inaccuracies, or refund delays.

Manually reviewing thousands of daily comments is unscalable. An automated, reliable NLP sentiment and topic classification system enables platforms to automatically route urgent complaints, flag struggling restaurant partners, and monitor delivery partner service levels in real time.

2.2 Project Objectives

The key objectives of the capstone project are:

- **Clean & Standardize Text:** Develop a robust preprocessing pipeline to strip noise while retaining sentiment-critical words like negations.
- **Feature Engineering:** Vectorize raw review text into numerical TF-IDF feature representations using unigrams and bigrams.
- **Supervised Classification:** Train and evaluate Multinomial Naive Bayes and Linear SVM models for sentiment classification.
- **Lexicon Baseline Comparison:** Benchmark machine learning models against VADER (Valence Aware Dictionary and sEntiment Reasoner).
- **Explore Dissatisfaction Themes:** Analyze frequent terms occurring in negative reviews to identify operational complaint themes.
- **Critical Evaluation & Methodological Rigor:** Audit data quality, assess dataset duplication impacts, and provide evidence-backed recommendations for real-world deployment.

3. Dataset Overview and Data Quality

3.1 Data Schema and Summary

The project utilizes a synthetic review dataset created specifically for learning and demonstration. It must not be represented as real Uber Eats operational data.

- **Original Dataset Path:** `data/raw/reviews.csv`
- **Total Row Count:** 1,984 records
- **Columns Available:**
  - `review_id`: Unique identifier for each review instance.
  - `order_id`: Associated customer order ID.
  - `customer_id`: Unique identifier for the customer.
  - `restaurant_id`: Unique identifier for the restaurant partner.
  - `review_text`: Raw customer feedback comment string.
  - `rating`: Integer star rating assigned by the user (1 to 5 stars).
  - `review_date`: Timestamp of review submission.
- **Missing Values:** 0 missing values across all required columns.

3.2 Duplicate Text Analysis & Template Repetition

A detailed audit of `data/processed/review_template_quality_check.csv` revealed a critical structural property of the dataset:

- **Total Rows:** 1,984
- **Unique Review Texts:** Exactly 15 unique text strings!
- **Template Frequency Examples:**
  - *"Excellent food and service"* — 301 occurrences
  - *"Fast delivery and delicious food"* — 273 occurrences
  - *"Food was cold and driver was late"* — 189 occurrences
  - *"Order was incomplete and food was stale"* — 164 occurrences

3.3 Template Quality & Rating Consistency

The `review_template_quality_check.csv` output confirmed that each of the 15 unique review templates maps consistently to exactly **one** star rating and one sentiment label (e.g., "Excellent food and service" is always 5 stars / Positive; "Food was cold and driver was late" is always 1 star / Negative).

While this eliminates conflicting ratings for identical text, it creates a synthetic environment where review text is perfectly deterministic of the assigned target label.

4. Sentiment Label Creation

4.1 Rating-Derived Proxy Rules

In the absence of manual human sentiment annotations, target sentiment labels were created directly from customer star ratings using standard proxy thresholds:

- **Ratings 1 – 2:** Labeled as `Negative`
- **Rating 3:** Labeled as `Neutral`
- **Ratings 4 – 5:** Labeled as `Positive`

4.2 Proxy Labels vs. True Sentiment Annotations

It is critical to distinguish **rating-derived proxy labels** from **independent human-annotated sentiment labels**:

- **Star Ratings as Proxies:** A customer might give a 3-star rating with text saying *"Food was amazing, but delivery took 2 hours!"* The star rating reflects an overall aggregated experience, whereas the text contains mixed or strongly negative delivery sentiment.
- **Evaluation Impact:** When a classifier is trained on star-derived labels, it learns to predict the overall rating tier rather than subtle textual sentiment nuances. Performance metrics evaluate alignment with the star rating rule, not necessarily human sentiment perception.

5. Text Preprocessing

To transform messy unstructured review text into clean tokens suitable for vectorization, the following multi-stage pipeline was implemented:

1. **Lowercasing:** All text was converted to lower case to ensure consistent representation (e.g., "Cold" and "cold" map to the same token).
2. **Noise & URL Removal:** URLs, web links, punctuation marks, special characters, digits, and emojis were stripped using regular expressions.
3. **Tokenization:** Text strings were broken into individual word tokens.

4. **Stop-Word Removal with Negation Preservation:** Standard NLTK English stop words (e.g., "the", "is", "at") were removed to reduce noise. However, **negation words**—specifically `"not"`, `"no"`, and `"nor"`—were explicitly preserved in the custom stop-word list.
5. **Lemmatization:** WordNet Lemmatizer was applied to reduce words to their base dictionary roots (e.g., "deliveries" → "delivery", "stale" → "stale").

5.1 Before-and-After Cleaning Example

- **Raw Review Text:** `"The delivery was NOT fast at all! Driver was very late and food was cold... http://ubereats.com/help"`
- **After Pipeline Processing:** `"delivery not fast driver late food cold"`

Why Negation Preservation Matters: Stripping the word "not" transforms `"not fast"` into `"fast"`, completely reversing the sentiment polarity from Negative to Positive. Preserving negations is essential for accurate sentiment analysis.

6. TF-IDF Feature Extraction

6.1 TF-IDF Concept & Mathematical Intuition

Term Frequency–Inverse Document Frequency (TF-IDF) converts unstructured text tokens into numerical feature vectors by scoring each word's relative importance across the dataset:

- **Term Frequency (TF):** Measures how frequently term t appears in document d.
- **Inverse Document Frequency (IDF):** Penalizes words that appear frequently across all documents (like "food" or "order"), boosting rare descriptive terms that carry high discriminative information.

6.2 N-Gram Configuration & High TF-IDF Terms

The vectorizer was configured with `ngram_range=(1, 2)` to extract both unigrams (single words like `"cold"`) and bigrams (two-word phrases like "cold food", "driver late"). Bigrams capture contextual phrases that single words miss.

Based on `data/processed/top_tfidf_terms.csv`, the highest weighted features across the vocabulary include:

1. `cold food` (Bigram)
2. `driver late` (Bigram)
3. `fast delivery` (Bigram)
4. `delicious food` (Bigram)
5. `missing item` (Bigram)

Note on Interpretation: A high TF-IDF score indicates that a term is highly descriptive and discriminative within specific documents; it does not automatically dictate whether the word is positive or negative until evaluated in model pipelines or class-based subsets.

7. Machine Learning Models

Two complementary supervised classifiers were implemented:

7.1 Multinomial Naive Bayes

- **Working Principle:** A probabilistic classifier based on Bayes' Theorem with the naive assumption of conditional independence between features given the class label:

- **Advantages:** Extremely fast to train, computationally lightweight, handles high-dimensional sparse text vectors remarkably well, and provides baseline probabilistic outputs.

- **Limitations:** The strong feature-independence assumption breaks down when complex word dependencies or long-range contextual relationships exist.

7.2 Linear Support Vector Machine (Linear SVM)

- **Working Principle:** A linear model that identifies the optimal decision hyperplane maximizing the margin of separation between sentiment classes in high-dimensional TF-IDF feature space.
- **Advantages:** Exceptionally robust in high-dimensional sparse spaces, effective against overfitting in text classification, and handles non-linearly separable boundary projections well.
- **Limitations:** Requires careful margin tuning (C parameter) and can be sensitive to extreme noise or mislabeled outliers.

7.3 Train/Test Split & Data Leakage Prevention

- **Split Ratio:** 75% Training set / 25% Test set (stratified split).
- **Pipeline Integrity:** The TF-IDF Vectorizer was fitted **strictly on the training set** (`fit_transform`) and subsequently applied to the test set using `transform` alone. This prevents vocabulary or IDF statistics from the test set leaking into model training.

8. Evaluation Methodology and Results

8.1 Comprehensive Evaluation Results Table

Source: `data/processed/model_evaluation.csv` & `data/processed/sentiment_method_comparison.csv`

Model / Approach	Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Evaluation Assessment
Multinomial Naive Bayes	**1.000000**	**1.000000**	**1.000000**	**1.000000**	Perfect recognition on duplicate test samples.
Linear SVM	**1.000000**	**1.000000**	**1.000000**	**1.000000**	Perfect linear separation on duplicate test samples.
VADER (Rule-Based)	**0.737903**	**0.537504**	**0.513031**	**0.519525**	Baseline rule accuracy against star-rating proxy labels.

9. VADER Sentiment Analysis

9.1 VADER Architecture & Compound Thresholds

VADER (Valence Aware Dictionary and sEntiment Reasoner) is an un-trained, rule-based sentiment analysis engine tailored for social media text, short reviews, emojis, and informal syntax. It computes a normalized Compound Score ranging from -1.0 (extreme negative) to $+1.0$ (extreme positive).

Standard threshold rules applied in the notebook:

- **Compound Score $\ge +0.05$:** Labeled as `Positive`
- **Compound Score ≤ -0.05:** Labeled as `Negative`
- **$-0.05 < \text{Compound Score} < +0.05$:** Labeled as `Neutral`

9.2 VADER vs. Trained ML Classifiers Comparison

- **Overall Accuracy:** VADER achieved **73.79%** accuracy versus **100.0%** for Naive Bayes and Linear SVM.
- **Macro F1-Score:** VADER scored **51.95%** versus **100.0%** for ML models.
- **Why Performance Differs:**

a. **Lexicon vs. Proxy Alignment:** VADER evaluates sentiment using pre-built lexicon dictionaries without training on the star ratings. When star ratings assign a 3-star rating as "Neutral", VADER may evaluate text like *"Good food but slow"* as slightly positive or negative based on word valence scores.
b. **Domain Specificity:** VADER's rule set does not natively recognize food-delivery domain phrases like *"driver dropped off order at wrong door"* as strongly negative unless explicit sentiment words like "bad" or "horrible" are present.

10. Complaint-Term and Customer Feedback Analysis

To discover key operational pain points, frequent terms occurring in negative review text (ratings 1–2) were extracted and analyzed (`data/processed/complaint_themes.csv`).

10.1 Observed Term Frequencies & Groupings

The top complaint n-grams extracted from negative reviews cluster into three main operational themes:

1. **Theme 1: Delivery Speed & Transit Time**
    - *Observed Terms:* `"late delivery"`, `"driver late"`, `"cold food"`, `"took too long"`.
    - *Text Evidence:* Reviews explicitly cite long wait times resulting in poor food temperature upon arrival.
2. **Theme 2: Order Accuracy & Completeness**
    - *Observed Terms:* `"missing item"`, `"wrong order"`, `"incomplete order"`, `"stale food"`.
    - *Text Evidence:* Customers report missing items or incorrect order items delivered by drivers.
3. **Theme 3: Packaging & Service Quality**
    - *Observed Terms:* `"bad service"`, `"damaged package"`, `"poor driver communication"`.

10.2 Important Methodological Distinction

- **Observed Evidence:** Specific negative review templates contain phrases like `"cold food"` and `"missing item"`.
- **Possible Interpretation:** Late transit times cause food cooling; restaurant prep errors cause missing items.
- **Action Requiring Validation:** Operational intervention should not occur based on keyword frequencies alone. Keywords identify *what customers write about*, but operational telemetry (GPS transit logs, kitchen prep timestamps, order checklists) is required to verify the root cause.

11. Visualizations and Interpretation

The analysis pipeline generated five core visualizations (`reports/figures/`), detailed below with their technical interpretation and business relevance:

11.1 Sentiment Distribution (`reports/figures/sentiment_distribution.png`)

- **What it shows:** Bar chart illustrating the total volume of reviews across Positive, Neutral, and Negative rating-derived categories.
- **Main Observation:** Positive reviews constitute the largest portion (~60%), followed by Negative reviews (~30%) and Neutral reviews (~10%).
- **Business Relevance:** Establishes the baseline customer satisfaction mix and confirms class imbalance levels for model training.

11.2 Naive Bayes & Linear SVM Confusion Matrices (`reports/figures/*_confusion_matrix.png`)

- **What it shows:** Heatmap representations of predicted vs. actual sentiment labels on the 496-row test set.
- **Main Observation:** 100% of test instances lie along the true diagonal, showing zero classification errors.
- **Business Relevance:** Visually proves perfect test performance while highlighting the need for validation on non-duplicate text.

11.3 Sentiment Method Comparison (`reports/figures/sentiment_method_comparison.png`)

- **What it shows:** Grouped bar chart comparing Accuracy, Precision, Recall, and F1-Score across Naive Bayes, Linear SVM, and VADER.
- **Main Observation:** ML models achieve 1.00 across all metrics, while VADER exhibits lower Macro Precision (0.54) and Recall (0.51).
- **Business Relevance:** Demonstrates the trade-off between out-of-the-box rule-based systems and domain-trained classifiers.

11.4 Top TF-IDF Terms & Complaint Terms (`reports/figures/top_tfidf_terms.png`, `reports/figures/complaint_terms.png`)

- **What it shows:** Horizontal bar charts of top TF-IDF feature weights overall and top term frequencies within negative reviews.
- **Main Observation:** Phrases like `"cold food"`, `"driver late"`, and `"missing item"` dominate the negative review feature space.
- **Business Relevance:** Directs operational focus toward delivery speed and order verification at partner kitchens.

11.5 Average Star Rating by VADER Sentiment (`reports/figures/average_rating_by_vader_sentiment.png`)

- **What it shows:** Bar chart displaying the mean star rating for reviews categorized as Negative, Neutral, and Positive by VADER.
- **Main Observation:** VADER Positive reviews average ~4.5 stars, Neutral reviews average ~3.0 stars, and Negative reviews average ~1.5 stars.
- **Business Relevance:** Validates that VADER's continuous compound scores correlate strongly with star rating trends.

12. Business Insights and Recommendations

Based on the empirical findings, the following evidence-backed, actionable recommendations are proposed:

1. **Diversify Review Data Collection (Immediate Technical Priority):**
    - *Action:* Expand data generation or real-world review collection to capture varied sentence structures, slang, typos, and long-tail feedback beyond 15 repeated templates.
2. **Target Operational Drivers of Dissatisfaction:**
    - *Action:* Initiate targeted quality audits on delivery routes and partner kitchens associated with frequent `"cold food"` and `"missing item"` terms.
3. **Implement Human-Annotated Evaluation Sets:**
    - *Action:* Annotate a gold-standard benchmark set of 500+ real customer comments to evaluate model accuracy against true human sentiment rather than star-rating proxies.
4. **Deploy Hybrid Sentiment Architecture:**
    - *Action:* Combine Linear SVM (for fast, high-accuracy category classification) with VADER/LLM fallback for low-confidence or sarcastic reviews.
5. **Cross-Validate Sentiment Signals with Operational Telemetry:**
    - *Action:* Link negative review spikes with kitchen prep time data and delivery GPS logs before initiating penalties or refund automation.

13. Limitations and Reliability Analysis

This section outlines the primary limitations and risks that must be acknowledged prior to real-world deployment:

13.1 Root Cause of 100% Test Accuracy (Data Leakage Artifact)

- **Dataset Structure:** 1,984 total rows containing only 15 unique review text templates.

- **Test Set Audit:** An audit of `data/processed/cleaned_reviews_deduplicated.csv` confirmed that **496 out of 496 test set rows (100%)** had cleaned text that was already present in the 1,488-row training set!
- **Impact:** The random row-level train/test split caused duplicate text templates to appear in both splits. The models did not learn generalized language patterns; they simply memorized that template A maps to label X. **100% accuracy must not be claimed as real-world predictive capability.**

14. Conclusion and Learning Outcomes

14.1 Project Conclusion

Day 3 successfully demonstrated an end-to-end NLP workflow for customer feedback analysis in a food-delivery context. Text preprocessing, TF-IDF feature extraction, supervised machine learning, and rule-based VADER sentiment analysis were fully implemented and evaluated. While Multinomial Naive Bayes and Linear SVM yielded 100% metric scores on the current dataset, rigorous data auditing revealed that this performance is driven by template duplication across train and test splits.

15. Technical Appendix

15.1 Environment & Python Libraries

- **Language & Runtime:** Python 3.10+
- **Core Libraries Used:**
  - `pandas`, `numpy`: Data manipulation, CSV parsing, and aggregation.
  - `scikit-learn`: `TfidfVectorizer`, `MultinomialNB`, `LinearSVC`, `train_test_split`, `classification_report`, `confusion_matrix`.
  - `nltk`: `word_tokenize`, `stopwords`, `WordNetLemmatizer`.
  - `vaderSentiment`: `SentimentIntensityAnalyzer`.

- `matplotlib`, `seaborn`: Visualization and confusion matrix heatmaps.

15.2 Output CSV Files & Artifacts

Filename / Path	Description & Purpose
`data/raw/reviews.csv`	Original raw synthetic review dataset (1,984 rows).
`data/processed/cleaned_reviews.csv`	Preprocessed review text with VADER sentiment scores.
`data/processed/cleaned_reviews_deduplicated.csv`	Deduplicated review dataset by unique review text.
`data/processed/review_template_quality_check.csv`	Quality check table mapping templates to rating consistency.
`data/processed/model_evaluation.csv`	Final metric table for Naive Bayes and Linear SVM models.
`data/processed/sentiment_method_comparison.csv`	Benchmark table comparing ML models against VADER.
`data/processed/top_tfidf_terms.csv`	Highest weighted TF-IDF unigram and bigram features.
`data/processed/complaint_themes.csv`	Frequent terms extracted from negative rating reviews.

15.3 Generated Visualization Figures

- `reports/figures/naive_bayes_confusion_matrix.png`
- `reports/figures/linear_svm_confusion_matrix.png`

- reports/figures/sentiment_method_comparison.png
- reports/figures/sentiment_distribution.png
- reports/figures/top_tfidf_terms.png
- reports/figures/complaint_terms.png
- reports/figures/average_rating_by_vader_sentiment.png